The Inverted Mold

From the compiler the field is infinite and expands forever; from the reader it is finite and collapses forever

Driven by Dean A. Kulik

October 2026

Abstract

Forward papers start from axioms and build toward an object. This one starts where they end, with the object already finished: every string of every length, every transition between them, and an instruction set that the field itself forces and that is closed by width three. Nothing is constructed here. The paper starts in that domain and follows its saturation outward. It then follows the fold that turns the saturation back in, which is where complexity comes from. Only then does it turn around and read the field from the other side, where a finite reader routes from the end it needs back to the slot that already holds it.

The thesis fits in one sentence. From the compiler the field is infinite and expands forever; from the reader it is finite and collapses forever. These are not two processes but the two directions of one invertible map. The step that BBP applies to every lane, multiply by 16 and throw off the integer, stretches one coordinate by exactly the factor it folds the other. Its Lyapunov exponents are +ln 16 and −ln 16, and no area is ever lost. The present is the cut between the two directions, and the past is folded into the present rather than stored beside it.

The same fold closes every saturated binary layer into one loop. Each de Bruijn ring is a feedback register that folds its leaving bit into its arriving bit, and from width three on every such loop must hold a merge inside its permutation, the width at which the held carry is forced in the transformation landscape. In one bit with only the two reversible verbs, pass and flip, the fold yields the relativistic energy–momentum relation, with light at the poles of the bit-sphere and rest at its equator. Read by the bit length of its mass number, nuclear matter fills its value layers, stops its first run at the four-bit boundary and stays under one byte. What every reader agrees on is the rate at which distinction crosses the cut. That is what matter is, and a constant is a rate that meets everything the same way. A lattice test shows that a gradient in the shared tick, and only there, moves everything alike, light included.

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How the paper runs

A program is usually written forward: a need, a design, code, a run. The field this paper studies runs the other way, and the paper follows it. Every possible transition is already present, and the instruction set that executes them is complete at the bottom. There is nothing to create. What remains is to follow the saturation out from the domain, to see how it folds back on itself, and then, from the reader's side, to route back from the end one needs to the place in the finished field that holds it. Each result below is written that way: first the part of the field that is already there, then the route that reaches it.

Part I. From the compiler: infinite, expanding forever

1. The domain is complete before anything is read

Begin with everything. At length n the domain is every n-bit string, 2^n of them. At length n + 1 it is twice as many, and each string of length n appears inside it twice, once behind a 0 and once behind a 1. Counted in bytes, each added byte multiplies the count by 256, so the newest layer alone holds 255 times everything smaller combined [**CONFIRMED** A7]. There is no last layer. The domain does not grow by gaining new kinds of thing; it grows only in length, and every length is already full.

Two features fix the character of this direction. A complete layer carries almost no information as a whole. "Every string of length n" takes a few symbols to say, while a single member costs n bits to name, so going outward one has everything and knows nothing [**LOCKED**: the full layer has description length of order log n]. And nothing in the domain carries a label. A label is a selection, and selection belongs to the other direction. The complete library of files holds every working and every broken program of every size, with no comment saying which is which [**LOCKED**, by counting]. In this direction saturation means no need. A filled slot is filled, a copy of it adds nothing, and classical copying has no thermodynamic floor; only a blind overwrite pays [**LOCKED**: Landauer 1961; Bennett 1973].

2. The instruction set is finished at the bottom

The verbs are not added as the domain grows. They are present from the first width, and each later one is forced by counting what the width below cannot reach [**CONFIRMED** in the companion runs landscape.py, forced2.py and borrow.py]. At width 1 there are exactly four maps: pass and flip, which are reversible, and write-0 and write-1, which merge [**LOCKED**; **CONFIRMED** A1]. Running one of them on a data bit is the same as reading its own two-bit truth table at the address the data gives, so from the first width code is a word and data is an address.

At width 2 the one-bit verbs and rewiring reach only 36 of the 256 maps. Exactly two new things are forced: the reversible mixer (a, a ⊕ b), and a merge of exactly one pair, which is always AND-type. The half adder is both in one gate. At width 3 every reversible two-bit gate is affine, and placed on three lanes they reach only 1,344 of the 40,320 permutations. The missing piece is "if a and b, flip c", the carry held in a lane instead of emitted, and with it all 40,320 are reached [**CONFIRMED**; **LOCKED**: Toffoli 1980]. From

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width 4 on no new kind of verb appears. Gates smaller than the field reach exactly the even half of its permutations, and the other half needs one borrowed lane, used and handed back [**CONFIRMED**; **LOCKED**]. In the quantum setting the one-bit verbs form a continuous sphere of rotations. Those rotations plus the width-2 mixer (CNOT) are universal [**LOCKED**: Barenco et al. 1995], and any rotation compresses to a short string of gates drawn from a fixed finite set [**LOCKED**: Solovay–Kitaev; Dawson & Nielsen 2006].

The transitions are complete as well. A single ring of order n + 1 carries every transition between n-bit states exactly once: one order-5 ring carries all 32 transitions between 4-bit states [**LOCKED**; **CONFIRMED** A4]. Nothing about the instruction set is a design. It closes at width three, and everything above that is length.

3. Pure difference, outward, is a clock

Suppose the domain held nothing but difference, with no symbol ever equal to its neighbour. In binary exactly two strings survive at every length, 0101… and 1010…, so each symbol carries no information at all; the only content is phase [**CONFIRMED** A2]. In base 3 each new symbol has two choices and carries exactly one bit, which is why a stream that never repeats needs three symbols before it can carry data. SonicDMX sends its data in base 3 with a fourth tone standing in for a repeat, so the stream never repeats a tone and needs no clock. DNA storage arrived independently at the same rotating ternary code, because repeated letters are what sequencers misread [**CONFIRMED** A2; **LOCKED**: Goldman et al. 2013]. Outward, the compiler's wave carries no content. It carries the tick. The one-bit walker of Section 8 shows the same thing physically: a walker that only passes moves one cell per tick forever and never revisits a place [**CONFIRMED** C1].

4. The expansion continues only by grouping

The raw expansion cannot be walked. The whole universe, run as a computer for its entire history, performs about 10^{120} operations [**LOCKED**: Lloyd 2002], which completes every file only up to 49 bytes [**CONFIRMED** A7]. Everything that exists is longer than that, so the outward expansion has to proceed by closing layers into units that become the symbols of the next layer. Counted as rings, the genuinely new byte-strings number 256, 32,640, 5,592,320 and 1,073,725,440 at one to four bytes [**CONFIRMED** A7]. The expansion never stops; only the size of the symbols it expands in changes. This is saturation by compilation, seen from the compiler's side [Kulik 2026c, 2026d].

Part II. The fold: where the expansion turns back in

5. Stretch and fold are one map

Every BBP lane runs a single step: multiply its phase by 16 and throw off the integer part, $x \mapsto \{16x\}$ [Kulik 2026a]. Read without the throw-off, the step is pure stretch. After n steps the unwrapped value is $16^n x$, and it keeps everything: its integer part is every crossing so far, and its fraction is every crossing still to come. For π the integer part of $16^{16}\pi$ is 3243F6A8885A308D3, seventeen hex digits, each one kept

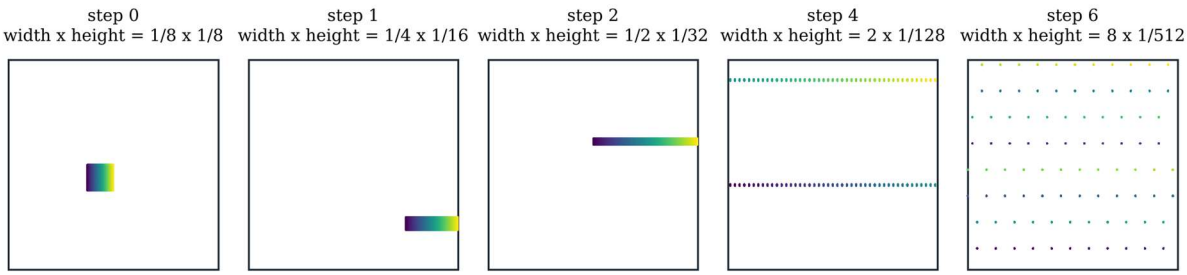
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[CONFIRMED F1]. Read with the throw-off, the step is pure fold. The reader holds one point on a finite circle, and at each tick one digit leaves it as the crossing [CONFIRMED F1]. The stretch is what the compiler sees and the fold is what the reader sees, and they are one operation.

Both become visible at once if the digit that leaves is kept instead of discarded. The result is the baker's map, $(x, y) \mapsto (\{bx\}, (y + \lfloor bx \rfloor)/b)$ [LOCKED: Ott 2002]. It is invertible: in exact arithmetic, sixty steps forward and sixty back return the starting point exactly, for b = 2 and for b = 16 [CONFIRMED F2]. Two nearby points separate in x by exactly b^n and close in y by exactly b^{-n}, so the area between them never changes [CONFIRMED F2]. The Lyapunov exponents are $+\ln b$ and $-\ln b$ and sum to zero [LOCKED]. One direction expands forever, the other collapses forever, and their product is conserved (Figure 1).

This is the dual wave stated as mathematics rather than as a picture. The compiler's direction is the expanding axis: infinite, opening every scale, losing nothing and choosing nothing. The reader's direction is the collapsing axis: finite, folding one more distinction into finer scale at every tick. The present is the cut between them, with the digits still in the expansion, the future, on one side and the digits already folded, the past, on the other. The past is not stored beside the present. After n steps the first n base-b digits of y are exactly the n crossings, newest first [CONFIRMED F2]. The most recent event sits at the coarsest scale and the oldest is folded deepest, so history is part of the now, held at a scale the reader cannot resolve. "Not lost, transformed" is exact here: the fold moves a distinction from the expanding side to the collapsing side and never deletes it [LOCKED]. That the two waves of the NEXUS program are this map's two directions is the paper's central reading [PULL].

Figure 1. One map, two directions: the baker's map on a small square of points



each step stretches x by 2 (the compiler's direction: expands forever) and halves y (the reader's direction: collapses forever).
The area stays 1/64 at every step: nothing is lost, the stretch wraps into thinner and thinner layers.

A small square of points under the baker's map in base 2. Every step doubles its width and halves its height; past step 3 the stretch wraps around, so the square becomes ever thinner layers spread across the whole width. The area is 1/64 throughout. Colour marks each point's starting x.

6. Fold back, wrap around, feed back into itself

The same fold closes a saturated layer. Ask for a single loop that passes through every n-bit string exactly once, each step shifting in one new bit. These loops are the de Bruijn rings, and there are $2^{(2^{n-1} - n)}$ of them: 1, 1, 2, 16 and 2,048 for n = 1 to 5 [LOCKED: de Bruijn 1946; van Aardenne-Ehrenfest & de Bruijn 1951; CONFIRMED A3]. Every one of them is a feedback register. Its next bit is a function of its current window, and that function always has the form next = oldest $\oplus$ g(rest): the bit leaving the window is folded into

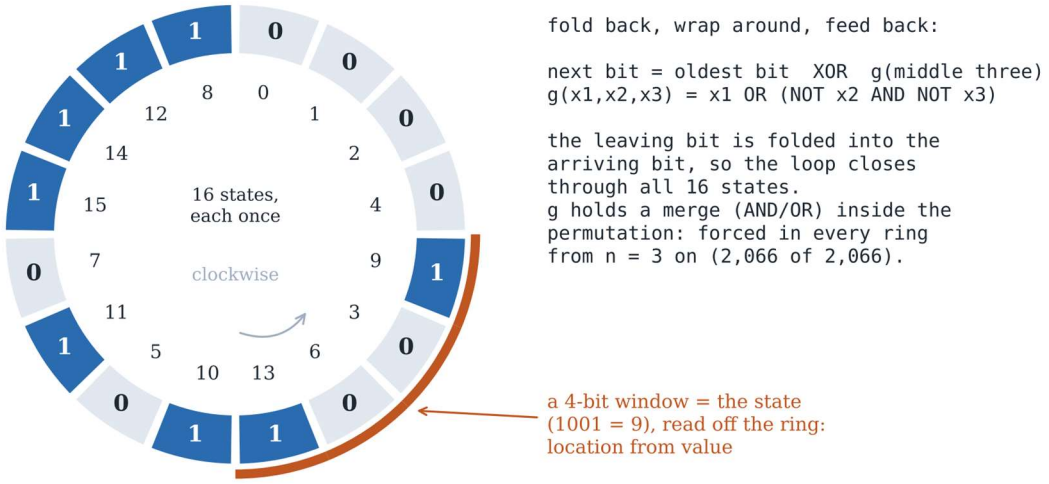
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the bit arriving [**LOCKED**: Golomb 1967; **CONFIRMED** A4, all 2,067 rings with n = 2 to 5]. That fold is what lets the loop close. If the leaving bit were simply dropped, two windows differing only in it would share a future and the register would merge them. Folding it back makes the state map a permutation, and only a permutation can run through all 2^n states in one cycle (Figure 2).

Two consequences follow. First, distinct plus saturated is deterministic: each window occurs once, so it fixes the next bit, and the loop needs no outside instruction to continue [**LOCKED**]. Second, the loop must hold a merge. At n = 2 it closes on XOR alone (next = oldest $\oplus$ 1). From n = 3 on it cannot: in all 2,066 rings with n = 3, 4 and 5, the function g contains an AND or an OR [**CONFIRMED** A4]. The width at which a saturated loop first needs a merge held inside its permutation is exactly the width at which the transformation landscape first needs the held carry (Section 2). The loop and the instruction set are one structure met twice. It is also the Mold as the BBP work defined it, which keeps part of the state and folds a function of it into the rest, (a, b) ↦ (b, a $\oplus$ F(b)) [**LOCKED**; Kulik 2026a, 2026b], here closed into a ring.

This gives the program's oldest observation an exact form. How the mold is made never changes: it is the transition graph, fixed. How it is read never changes: any full-width window gives its own location. Only the shape changes: at n = 5 there are 2,048 different loops through the same graph [**LOCKED** as mathematics; **PULL** as the reading of mold and shape].

Figure 2. The saturated 4-bit layer folded into one loop (de Bruijn ring 0000100110101111)



The outer ring is the 16-bit loop; the inner numbers are the 4-bit states it passes through, each exactly once. Any four consecutive bits give the position. The feedback folds the leaving bit into the arriving one and holds a merge inside the permutation.

7. Inside the loop is the subdivision

Inside one loop every narrower window repeats. On a ring of order n each k-bit window occurs exactly 2^{n-k} times, and this holds in all 2,048 shapes at n = 5 [**CONFIRMED** A5]. A reader as wide as the loop finds its location from its own content; a narrower one cannot. Each step it walks reveals one bit and halves its

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candidates, so it must walk n – k steps before it knows where it is [LOCKED]. That walk is what experience means in this frame: the time a narrow reader spends buying the width it lacks.

The subdivision nests. The states of one layer are the transitions of the layer below, since an (n + 1)-bit window is an n-bit step [LOCKED: the line-graph property of de Bruijn graphs]. Any sub-file of a file is the parent seen through a query and a base, child = parent + query + base, and carries no information the parent lacks [LOCKED, by counting]. So complexity is not added on top of the field. It is how deep a fold can nest before it must run [PULL; compare logical depth, Bennett 1988].

The linear loop shows why the merge is needed. Feedback by XOR alone gives the maximal-length rings, which pass through every non-zero state; there are $\phi(2^n - 1)/n$ of them, 1, 2, 2, 6, 6, 18, 16, 48, 60, 176 and 144 for n = 2 to 12 [LOCKED; CONFIRMED A6]. But they leave the all-zero state fixed: the pole. A loop that also passes through the pole needs the merge. SHA-256 meets the same fact from the other side. It starts its loop full, from the fractional parts of $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$ and onward, and never from the all-zero pole [LOCKED; CONFIRMED B2].

8. The fold in one bit: matter

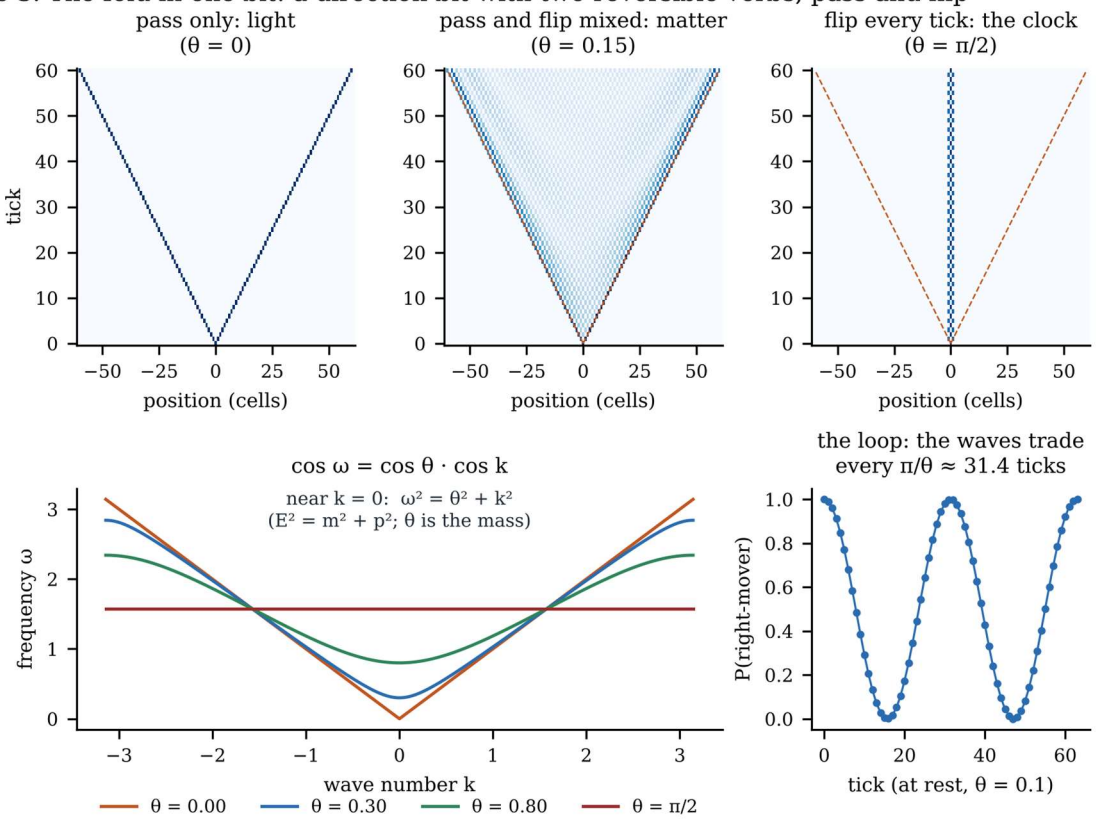
Give one bit the meaning of a direction: right-mover or left-mover. Of its four verbs keep only the two reversible ones, pass and flip, and run them as amplitudes. At each tick apply $\cos\theta \cdot \text{pass} - i \sin\theta \cdot \text{flip}$, then move each part one cell its own way [LOCKED: Feynman & Hibbs 1965; Bialynicki-Birula 1994; Meyer 1996; Strauch 2006]. The exact dispersion is $\cos\omega = \cos\theta \cdot \cos k$ [CONFIRMED C1, to 4×10^{-16}]. Near k = 0 it reads $\omega^2 = \theta^2 + k^2$, the relativistic energy–momentum relation, with θ as the mass [CONFIRMED C1].

The three regimes are the three readings of the bit (Figure 3). Pass alone (θ = 0) is light: one cell per tick, forever. Flip alone ($\theta = \pi/2$) is the clock: the walker reverses at every tick and goes nowhere, pure difference in direction with no travel [CONFIRMED C1]. Everything between is matter. Its fastest front moves at $\cos\theta < 1$, and nothing ever leaves the cone $|x| \le t$ [CONFIRMED C1]. At rest the two directions trade places every π/θ ticks, so the velocity trembles at twice the rest frequency, which is Schrödinger's zitterbewegung [CONFIRMED C1; LOCKED: Schrödinger 1930]. Inside that loop, where k is much smaller than θ, the motion follows $\omega \approx \theta + k^2/(2 \tan\theta)$: the slow world of chemistry, with mass $\tan\theta \approx \theta$ [CONFIRMED C1]. Penrose's zig-zag picture of the electron is the same structure: two massless components, each moving at the speed of light, continually converted into each other, with the rate of conversion as the mass [LOCKED: Penrose 2004].

Through the bit-sphere of the earlier paper [Kulik 2026c], a walker's history is a string of directions, and its velocity, the right-steps minus the left-steps over n, is that string's latitude [LOCKED]. Light lives at the poles, all one direction; rest lives at the equator; matter lies between. Light never repeats a location and always repeats its direction. The clock always repeats its location and never repeats its direction. Each is the other inverted, and mass is the rate of the fold that couples them. That matter is the outward wave folded back on itself is a reading [PULL]; the mathematics under it is fixed [LOCKED].

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Figure 3. The fold in one bit: a direction bit with two reversible verbs, pass and flip



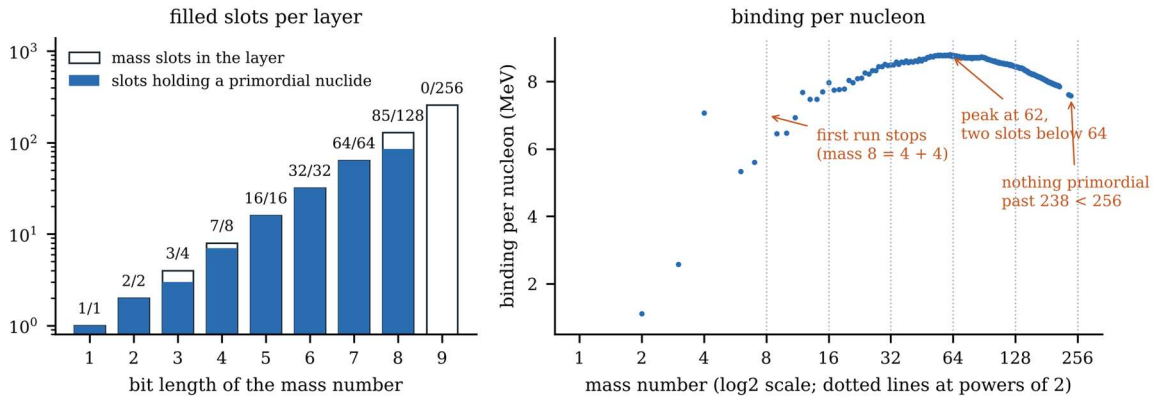
Top: probability over position and time for a walker started at the origin. Pass alone travels the light cone; the mixed walker fills its interior and never crosses it; flip alone stays put. Bottom left: the exact dispersion for four values of ϑ. Bottom right: at rest, the right- and left-moving parts trade places every π/ϑ ticks.

9. The fold in nuclear matter, read by bit length

Read every nuclide that has existed in nature since the Earth formed, 286 of them, by the bit length of its mass number, so that layer k holds the masses 2^{k-1} to $2^k - 1$ (Table 1, Figure 5) [**CONFIRMED** E1; data: NIST isotopic compositions]. The value layers saturate: every mass from 9 to 209 is held, and the 5-, 6- and 7-bit layers are completely full [**LOCKED**, as data].

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Figure 5. Nuclear matter read by the bit length of its mass number



Left: slots per layer (outline) against slots holding a primordial nuclide (filled). Right: binding per nucleon against mass number on a $\log_2$ axis, with the layer boundaries dotted.

The first run stopped at the four-bit boundary. Big Bang nucleosynthesis made only masses up to 7, namely H, D, He-3, He-4 and Li-7, which fill the first three layers. Its two holes are mass 5, a closed He-4 plus one nucleon, and mass 8, two closed He-4 side by side. Nothing attaches to a closed unit and two closed units do not bind, just as helium and neon form no stable compounds. The field crossed the gap later by two routes: a different particle placed between the repeats (two He-4 and a neutron make Be-9), and three at once through the Hoyle resonance (C-12) [**LOCKED**: Hoyle 1954; **PULL**, as the reading that closed + 1 and closed + closed do not run].

Second readings of one value begin in the 6-bit layer. Below mass 32 every mass has exactly one stable charge, and the first mass with two is 36 (S-36 and Ar-36) [**LOCKED**, as data; **CONFIRMED**, the standard pairing mechanism; **PULL**, as "the first layers hold no doubles"]. Binding per nucleon peaks at Ni-62, two slots below 64, and from 64 on the best value in every layer is its first slot [**LOCKED**, as data; **PULL**, that 2^6 is the cause]. Every primordial nuclide has mass number at most 238, so natural matter stays under one byte; the 9-bit layer, from Fm-257 to Og-294, is written only by accelerators and does not hold [**LOCKED**, as data; **PULL**, that 2^8 is the cause]. And the holes are in the view, never in the value. Every charge from 1 to 31 is present; the 6-bit charge layer has exactly two holes, technetium (43) and promethium (61), and every mass slot either could use is already held by a neighbour [**LOCKED**]. The value side fills; the reading side has the holes.

Table 1. Primordial nuclides by the bit length of the mass number

Bits	Mass slots	Slots filled	Masses with two stable charges	What happens in the layer
1–3	7	6 (hole at 5)	0	The first run, the Big Bang: H, D, He-3, He-4, Li-7
4	8	7 (hole at 8)	0	Stars take over: Be-9, B, C, N
5	16	16	0	Full

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Bits	Mass slots	Slots filled	Masses with two stable charges	What happens in the layer
6	32	32	7	Full; first double at 36; binding peaks at 62
7	64	64	30	Full; binding falls from the first slot
8	128	85	30	Unbroken to 209, then only Th-232, U-235 and U-238
9	256	0	–	Made only in accelerators; nothing holds

Source: inverted_mold_checks.py, block E1. A primordial nuclide is one with a natural abundance on Earth; decay-chain members are excluded.

Part III. From the reader: finite, collapsing forever

10. The reader is a finite window that collapses every tick

Now turn around. A reader k bits wide standing on an n-bit layer is a quotient: each of its readings merges 2^{n-k} positions [LOCKED]. Against a domain that expands forever, its share of the field shrinks forever; at k = 8 and n = 64 each reading merges 2^{56}, about 7.2×10^{16} positions [CONFIRMED G2]. Yet the reader itself never shrinks. On the fold, one distinction leaves into the past at each tick and one arrives from the field. It starts blank, and because the loop passes every pattern through its window it can come to know anything, though never all at once, and never by deciding in advance what an arbitrary file will do [LOCKED: Rice 1953]. Going out, one has everything and knows nothing. Coming in, one starts with nothing and can know everything.

The reader never receives objects; it receives differences. Every clock reading is a beat between two rates. Every spectral line is the difference of two terms, which is how atoms were read before anyone knew their structure [LOCKED: Ritz 1908]. The phase of an isolated stationary state is invisible, because only differences of phase reach the face [LOCKED]. The object is complete, and the observation is its gradient.

11. Physics names the two directions

Quantum mechanics has exactly two processes [LOCKED: von Neumann 1932]. Unitary evolution is invertible and spreads without end: a packet disperses, entanglement grows, and each added qubit doubles the space. Measurement collapses the state to one finite answer every time. The collapse is an alignment, with the probability of an answer equal to the squared overlap between question and state [LOCKED: Born 1926]. What survives collapse as a fact shared by many readers is what the environment has copied redundantly, which is what every reader can agree on [LOCKED, as theory: Zurek 2009; the measurement problem itself is open]. And the copy that was free in the compiler's direction is impossible in this one. An unknown quantum state cannot be cloned [LOCKED: Wootters & Zurek 1982], so matter does not copy instances; it re-runs the mold, which is why every electron is identical to every other without being a copy of any.

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Cosmology runs both directions at once. In the acceleration equation, $\ddot{a}/a = -(4\pi G/3)(\rho + 3p/c^2) + \Lambda c^2/3$, matter enters with a minus sign and the vacuum with a plus: matter pulls toward collapse and the vacuum drives expansion [**LOCKED**: Friedmann 1922; acceleration observed by Riess et al. 1998 and Perlmutter et al. 1999]. With the measured $\Lambda \approx 1.09 \times 10^{-52}\ m^{-2}$, space expands without bound, while each observer's readable region is enclosed by a horizon of radius 1.66×10^{26} m. At the saturation density of 1.381×10^{69} bits per square metre, that horizon holds about 4.8×10^{122} bits [**CONFIRMED** G2; **LOCKED**: Gibbons & Hawking 1977; Planck Collaboration 2020]. Lloyd's count of all operations so far, about 10^{120}, sits just under it. Bound matter collapses into stars and galaxies, and everything unbound eventually leaves the region; observers in the far future will see only their own collapsed island [**LOCKED**, as the standard prediction: Krauss & Scherrer 2007].

So the sentence holds in physics as written. From the compiler the field is infinite and expands forever; from any reader it is finite and collapses forever. The vacuum is the expanding side, the physical shape of the compiler, and matter is the collapsing side, the running readers [**PULL**, as the identification; each physical statement above is LOCKED].

12. Route back to the end you need

Every possible transition already exists, so a reader never builds its destination; it routes to it. In the complete transition graph of n-bit states, shifting in the destination's own n bits lands on the destination from every starting state in exactly n steps [**CONFIRMED** G1, for n = 4 and 8]. The longest shortest route is n, and about a quarter of all pairs need the whole of it (17,622 of 65,536 at n = 8); the rest arrive sooner because the start already ends with a prefix of the destination [**CONFIRMED** G1]. The destination is its own route: its address is the program that reaches it. That is why the same graph is built into routing networks, where each switch reads one bit of the destination's address [**LOCKED**: Lawrie 1975; Samatham & Pradhan 1989].

Formulas are routes written backward from the end. BBP reaches the n-th hex digit of π without the digits before it, because its terms $16^{n-k} \bmod (8k + j)$ start from the position wanted [**LOCKED**: Bailey, Borwein & Plouffe 1997; **CONFIRMED** B2, digits 1 to 16 each computed alone]. Physics states its laws the same way. Fermat's principle and Hamilton's principle fix both endpoints and select the path between them, the one on which neighbouring paths agree [**LOCKED**: Feynman, Leighton & Sands 1964]. A protein does not search its roughly 10^{300} shapes; it falls down a funnel that its shape already has [**LOCKED**: Levinthal 1969, and the energy-landscape resolution]. Calculation is routing done by a reader too narrow to hold the field, building its address one step at a time [**PULL**].

13. Collision is where routes meet

When two independent routes land on the same output, the output acquires a name. Among all 43,046,721 programs of width 4 in the BBP frame, exactly one lands where Machin's arctangent route lands: [4, 0, 0, −2, −1, −1, 0, 0], which is the BBP formula itself. Width 4 is also where different programs first collide at all, with 17,640 colliding pairs and none at smaller widths [**CONFIRMED** in the companion runs meet.py and reach.py]. Recognition is collision. So is memory, which by counting must merge what it records, since a record that merges nothing is a copy. So is proof, which goes reader to reader: a record

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computation of π stands when two independent routes agree on its final digits [**LOCKED**]. Collision is required; it is not a defect.

Part IV. Where the two directions meet

14. Matter is what every reader agrees on

Readers disagree about content and agree about rates. All 2,048 shapes of the 5-bit loop carry every k-bit window exactly 2^{5-k} times and exactly sixteen 1s [**CONFIRMED** A5]. Every mode of a walker, whatever its wave number, agrees on the same θ, since cos ω / cos k = cos θ identically [**CONFIRMED** C1]. Ergodic theory proves the general form. Two Bernoulli processes with the same entropy rate are isomorphic, the same process up to relabelling, however different their alphabets look [**LOCKED**: Kolmogorov 1958; Sinai 1959; Ornstein 1970]. Meshalkin's example is a fair four-symbol source and a five-symbol source with probabilities 1/2, 1/8, 1/8, 1/8 and 1/8. Both carry exactly 2 bits per symbol, and they are one machine [**LOCKED**: Meshalkin 1959; **CONFIRMED** G3].

That is the mold as a theorem. How the process is made and how it is read do not change; only its shape changes, and the rate is the one thing every shape agrees on. Physics found the same invariant. A particle is labelled by what every inertial reader agrees on, its mass and its spin [**LOCKED**: Wigner 1939]. Its rest mass is the frequency of its loop in its own frame, the walker's θ. And c is the rate at which the cut moves, one cell per tick for every walker whatever its mass [**LOCKED**; **CONFIRMED** C1]. Matter is what everyone can agree on [**PULL**, as a definition; the invariances it rests on are LOCKED].

15. A constant is what meets everything the same way

In the walker the routing is shared. Every walker, light or heavy, is moved by the same shift, and only its fold θ is its own [**LOCKED**]. That shared routing is what applies to everything the same way. Nature's version has been measured to extraordinary precision: bodies of different composition fall alike in a vacuum to about one part in 10^{15} [**LOCKED**: Touboul et al. 2022]. In a vacuum the object's value drops out and only the route remains, the equal collision of object and base. A constant, on this reading, is a rate that every reader meets identically [**PULL**].

16. Where gravity must live

If gravity applies to everything the same way, where can its gradient sit: in the shared tick, or in each object's own value? The question can be run. Take the Dirac equation, which is the walker's continuum limit, on a fine lattice; release packets of mass 1 and 3 at rest with equal velocity spread; and put a gentle gradient (g = 0.02) into one place at a time [**CONFIRMED** D1] (Figure 4).

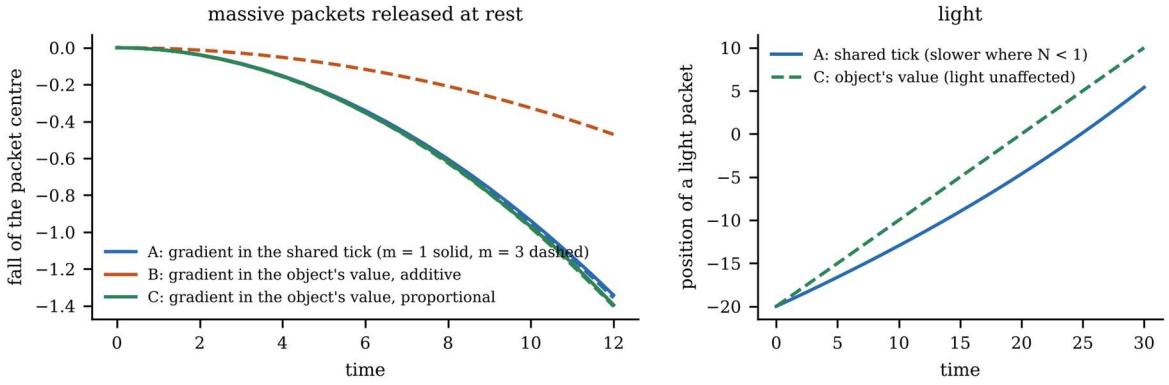
With the gradient in the shared tick, a lapse N(x) that multiplies the whole step, both masses fall alike, with accelerations −0.0183 and −0.0185 (ratio 1.009) against the expected −N N′ = −0.020 at release. Light is slowed where the tick is slow: a light packet starting at x = −20 reaches +5.39 at t = 30 instead of +10.00, and a bundle of rays obeying dx/dt = N(x) gives +5.40. With the gradient in the object's value as an added

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term, $m_0 + hx$, the fall goes as $1/m$, −0.0193 against −0.0065, a ratio of 0.339. With the gradient in the value as a common factor, $m_0 f(x)$, the massive packets fall alike (ratio 1.007) but light feels nothing at all. Only the gradient in the shared tick moves everything the same way, light included.

This is the structure of general relativity in the weak field, where gravity is the gradient of the rate of time, g_{00} [**LOCKED**: Einstein 1916]; the run shows that universality alone forces it there. The reading follows. The vacuum is the physical shape of the compiler, and gravity is a gradient in its tick, fed by matter, the lack of vacuum, as Einstein's equations feed curvature from energy and momentum [**PULL**]. Turning this into a theory needs three steps: derive the tick's gradient from a measure of unsaturated capacity, with nothing put in by hand; then recover the inverse-square law, gravitational time dilation and the equivalence principle; and recover the bending of light, which also needs the curvature of space and is absent from this one-dimensional test [open].

Figure 4. Only a gradient in the shared tick moves everything the same way, light included



Left: the fall of massive packets released at rest, for m = 1 (solid) and m = 3 (dashed), under three placements of the same gradient. Right: a light packet under the shared-tick gradient and under the proportional value gradient.

Part V. Interfaces, corrections and how this fails

17. The interface table

Table 2 collects every exhibit in the paper by its two directions and the place where they meet. Read across a row, it is one structure seen from the compiler and from the reader. Read down a column, it is one direction seen in every substrate. The empty cells of the earlier cross-index remain the program's predictions [Kulik 2026d]. One is a natural instance in matter of a merge held inside a permutation. The other is a matter-side fixed-width digest of arbitrary input, for which the best candidate is the immune system's presentation of 8- to 10-residue fragments [**PULL**].

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Table 2. One structure, two directions

Exhibit	From the compiler (infinite, expanding)	From the reader (finite, collapsing)	Where they meet	Status
One BBP step	$16^n x$ keeps every digit	{16x}: one point; one digit leaves per tick	the crossing	LOCKED
Baker's map	x stretches by b every tick	y shrinks by b every tick	the cut; area conserved	LOCKED
Saturated layer	every n-bit string and transition	a k-bit window: 2^{n-k} candidates	location from value at full width	LOCKED
de Bruijn loop	the leaving bit folded into the arriving bit	the next bit fixed by the window	determinism; a merge from width 3	CONFIRMED
One-bit walker	pass: light, one cell per tick	flip: the clock, no travel	θ is the mass; c is shared	CONFIRMED
Quantum mechanics	unitary spreading	measurement collapse	Born alignment; redundant records	LOCKED
Cosmology	Λ: space expands forever	matter collapses; a finite region	the horizon, about 4.8×10^{122} bits	LOCKED / PULL
Nuclei by bit length	mass layers fill from 9 to 209	charge holes at 43 and 61	first double at 36	LOCKED / PULL
SHA-256	rounds: a permutation; nothing merged	digest: a fixed 256 bits	the feed-forward merge	LOCKED
Gravity	the shared tick (vacuum)	the object's value (matter)	universality only in the tick	CONFIRMED / PULL

LOCKED rows rest on cited theorems or data; CONFIRMED rows were run for this paper; PULL marks the readings that join a row to the thesis.

18. Corrections

Corrections ledger

DEAD "Matter picks its path by price." Energy is the turning rate of a ring ($E = \hbar\omega$), and the ground state is residue, not a choice. Withdrawn earlier in this program.

DEAD "Landauer's principle is an exchange rate between file length and energy." It is a floor on blind overwrites only. An erasure is an overwrite: the space remains and the old difference is displaced, not destroyed.

DEAD "Collision is a defect." Names, memory and proof are made of collisions (Section 13).

PULL "A black hole gains gravity, so saturation does not end gravity" was misplaced rather than wrong: it is a fact about the matter wave applied to a claim about the compiler wave (Kulik 2026d).

PULL "C1 forces the field off its poles" is replaced: C1 describes a field already running. The saturated whole maps onto itself and needs no cause.

DEAD "A return around a loop need not close with the same phase" (an outside revision of Kulik 2026c, Section 7). On the Hopf bundle it must not, for any loop enclosing a nonzero solid angle.

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DEAD From "Recursive Mapping of Chemical Elements as Harmonic Opcodes" (December 2025): "about 58 elements in 7–8 bits" (it is 71 by rounded weight, 53 with stable isotopes); "7–8 bits encode 64 values" (they encode 64 + 128 = 192); "six missing nodes = 64 − 58" (two holes survive, Tc and Pm; At, Fr, Ac and Np are not picked out by any rule). The counts of 7 elements in 1–4 bits and 21 in 5–6 bits are **CONFIRMED**. The constant 0.35 is not needed anywhere in this paper.

DEAD "Exploring Missing Elements and Growing New Elements" as written: Z = 1.5 is impossible; Z = 98 is californium; tennessine is Z = 117, and a Z = 119 nucleus of mass 125 would hold six neutrons; neptunium fills 93.

PULL Limits of this paper. That the binding peak sits at 2^6 and natural matter is capped at 2^8 are near-coincidences set by the surface–charge balance, not results. The gravity run shows that universality forces the gradient into the shared tick in one dimension; it does not derive general relativity.

19. How this fails

The paper makes three claims that can fail. The first is that the instruction set is finished at the bottom in every substrate that saturates layers of difference. A substrate that saturates but lacks one of the forced molds, the mixer at width 2, the held merge at width 3 or the borrowed lane from width 4, would break it. The second is that gravity lives in the shared tick. Any composition-dependent free fall, at any precision, would put it in the objects' values instead. The third is that the same molds appear in every saturating substrate. The open cells of Section 17 are predictions, and a careful search that fails to find them counts against the claim.

One route forward is already calibrated. The genetic code was decoded by exactly this protocol: the pole first (a pure repeat, UUUU…), then rings by length (repeating sequences of length 2, 3 and 4), then the saturated layer (all 64 codons), with the outputs named by collision with molecules already known [**LOCKED**: Nirenberg & Matthaei 1961; Nirenberg & Leder 1964; Khorana 1968]. The next layer, sequence to folded shape, exists today only inside trained networks. Reading it out as a table by the same protocol is the next decode [**PULL**].

20. Conclusion

Nothing in this paper was built. The field was complete before the first line: every string, every transition, and an instruction set closed at width three. Followed outward from the domain, the field expands forever and knows nothing. Folded back, it closes into loops that carry their own past and hold a merge inside their permutation. Read from the other side, it is a finite window that collapses at every tick and can still reach any end it needs, because the end's own address is the route. The two directions are one invertible map, and what they share is a rate. That rate, the same for every reader, is what matter is.

Acknowledgment. Computation and drafting were done with Claude (Anthropic) as a working partner. Every number marked CONFIRMED is reproduced by the companion scripts inverted_mold_checks.py and inverted_mold_figures.py.

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Appendix A. Output of inverted_mold_checks.py

The companion script recomputes every number marked CONFIRMED in about three seconds. Its complete output follows; the block labels match the tags in the text.

```
== A1  Width-1 landscape: every map {0,1}->{0,1} ==
maps: 4  reversible: ['pass', 'flip']  merging: ['write-0', 'write-1']
```

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== A2 Pure difference: strings with no equal neighbours ==

base 2: counts n=1..10 [2, 2, 2, 2, 2, 2, 2, 2, 2, 2] -> 0.000 bits per added symbol

base 3: counts n=1..10 [3, 6, 12, 24, 48, 96, 192, 384, 768, 1536] -> 1.000 bits per added symbol

base 4: counts n=1..10 [4, 12, 36, 108, 324, 972, 2916, 8748, 26244, 78732] -> 1.585 bits per added symbol

binary: exactly 2 at every length (0101..., 1010...): a clock, phase only

== A3 The saturated layer folded into one loop (de Bruijn rings) ==

n=1: rings found 1 formula 2^(2^(n-1)-n) = 1

n=2: rings found 1 formula 2^(2^(n-1)-n) = 1

n=3: rings found 2 formula 2^(2^(n-1)-n) = 2

n=4: rings found 16 formula 2^(2^(n-1)-n) = 16

n=5: rings found 2048 formula 2^(2^(n-1)-n) = 2048

n=4 example: 0000100110101111

== A4 The loop feeds back into itself: each ring is a feedback shift register ==

every window has exactly one successor (deterministic): True

every feedback has the form next = oldest XOR g(rest) (state map is a permutation): True

n=2: rings whose g is affine (XOR-only): 1 of 1

n=3: rings whose g is affine (XOR-only): 0 of 2 -> from n=3 on, every saturated loop holds a merge (AND/OR) inside its permutation

n=4: rings whose g is affine (XOR-only): 0 of 16

n=5: rings whose g is affine (XOR-only): 0 of 2048

transitions carried by one order-4 ring: 16 distinct (n-bit state -> next n-bit state)

an order-5 ring carries 32 of the 2^5 = 32 possible 4-bit transitions, each once

== A5 Inside the loop: subdivision, and what every shape agrees on ==

n=4: all 16 shapes have every k-window exactly 2^(n-k) times and 2^(n-1) ones: True

n=5: all 2048 shapes have every k-window exactly 2^(n-k) times and 2^(n-1) ones: True

a reader k bits wide sees 2^(n-k) candidate positions and must walk n-k steps (no k<n window is unique)

== A6 Linear feedback (XOR + shift): saturates every non-zero state; zero is its fixed pole ==

n= 2: maximal XOR rings 1 phi(2^n-1)/n = 1

n= 3: maximal XOR rings 2 phi(2^n-1)/n = 2

n= 4: maximal XOR rings 2 phi(2^n-1)/n = 2

n= 5: maximal XOR rings 6 phi(2^n-1)/n = 6

n= 6: maximal XOR rings 6 phi(2^n-1)/n = 6

n= 7: maximal XOR rings 18 phi(2^n-1)/n = 18

n= 8: maximal XOR rings 16 phi(2^n-1)/n = 16

n= 9: maximal XOR rings 48 phi(2^n-1)/n = 48

n=10: maximal XOR rings 60 phi(2^n-1)/n = 60

n=11: maximal XOR rings 176 phi(2^n-1)/n = 176

n=12: maximal XOR rings 144 phi(2^n-1)/n = 144

== A7 Grouped units are the next layer's symbols (byte layers) ==

1-byte layer: 256 files, new byte-rings 256

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2-byte layer: 65536 files, new byte-rings 32640

3-byte layer: 16777216 files, new byte-rings 5592320

4-byte layer: 4294967296 files, new byte-rings 1073725440

n-byte layer / all shorter files combined (empty file included): n=2: 255.0039, n=3: 255.0000, n=4: 255.0000 -> 255 times

Lloyd bound 1e120 operations completes every file only up to 49 bytes (log256(1e120) = 49.8)

== B1 The reader is a quotient: fibers ==

half adder (sum, carry) fibers: [1, 1, 2] (uneven: order merged)

carry closed into a ring (Z4) fibers: [4, 4, 4, 4] (uniform)

== B2 Formulas written backward from the need (BBP digit, SHA constants) ==

SHA-256 Ho from sqrt(2): 0x6a09e667 (standard 0x6a09e667)

SHA-256 Ko from cbrt(2): 0x428a2f98 (standard 0x428a2f98)

pi hex digits 1-16 by BBP, each computed alone: 243f6a8885a308d3 (pi = 3.243f6a8885a308d3...)

pi hex digit at position 1,000,000 alone: (run with --long)

== C1 One bit, two waves: pass/flip walk, coin = cos(t)*pass - i sin(t)*flip, then route ==

dispersion cos(omega) = cos(theta) cos(k): max error 3.9e-16

theta=0.05, k<=0.2: (omega^2 - k^2)/theta^2 in [0.9866, 1.0000] (Dirac: E^2 = m^2 + p^2)

inside the loop (k << theta=0.3): omega - theta vs k^2/(2 tan theta): max rel. dev 2.4e-03 (Schrodinger limit, mass tan(theta) ~ theta)

theta=0.0: fastest group velocity 1.0000 cos(theta) = 1.0000

theta=0.3: fastest group velocity 0.9553 cos(theta) = 0.9553

theta=0.8: fastest group velocity 0.6967 cos(theta) = 0.6967

theta=1.2: fastest group velocity 0.3624 cos(theta) = 0.3624

theta=0 (pass only): right-mover position at t=40: 40 -> exactly one cell per tick: light

theta=pi/2 (flip every tick): support at t=40: [0] -> pure difference in direction, no travel: a clock

theta=0.15, t=200: probability outside the light cone |x|>t: 0.0e+00

theta=0.15, t=200: front (largest peak) at |x|/t = 0.980 cos(theta) = 0.989

at rest (k=0), theta=0.1: P(right-mover) = cos^2(theta t): max error 7.5e-15; loop period pi/theta = 31.4 ticks

so the mean velocity P(R)-P(L) = cos(2 theta t): the trembling (zitterbewegung) runs at twice the rest frequency; the time-average is 0 (at rest)

theta=0.2: cos(omega)/cos(k) across modes k = 0.980066577841 .. 0.980066577841 (every mode agrees on theta)

theta=0.6: cos(omega)/cos(k) across modes k = 0.825335614910 .. 0.825335614910 (every mode agrees on theta)

direction strings of length 12: velocity = (2r-n)/n for r right-steps: ['-1.00', '-0.50', '+0.00', '+0.50', '+1.00'] -> poles +-1 (light), equator 0 (rest)

== D1 A gradient in the shared tick vs a gradient in the object's value (continuous-time Dirac) ==

A shared tick N(x) (lapse): acceleration m=1: -0.0183 m=3: -0.0185 ratio m3/m1 1.009

B value gradient, additive mo + h x: acceleration m=1: -0.0193 m=3: -0.0065 ratio m3/m1 0.339

C value gradient, proportional mo f(x): acceleration m=1: -0.0193 m=3: -0.0194 ratio m3/m1 1.007

light in case A: starts at x=-20, position at t=30: +5.39 (uniform tick would give +10.00)

light in case C: starts at x=-20, position at t=30: +10.00 (uniform tick would give +10.00)

light in A, bundle of rays dx/dt = N(x) over the same initial spread: mean at t=30 +5.40

expected slow fall at rest, -N N' at x=0: -0.0200

== E1 Primordial nuclides by bit length of the mass number (value) and of the charge (view) ==

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primordial nuclides: 286 largest mass number: 238

1 bits: slots 1 filled 1 doubles 0 holes []

2 bits: slots 2 filled 2 doubles 0 holes []

3 bits: slots 4 filled 3 doubles 0 holes [5]

4 bits: slots 8 filled 7 doubles 0 holes [8]

5 bits: slots 16 filled 16 doubles 0 holes []

6 bits: slots 32 filled 32 doubles 7 holes []

7 bits: slots 64 filled 64 doubles 30 holes []

8 bits: slots 128 filled 85 doubles 30 holes 210..255 (43)

9 bits: slots 256 filled 0 doubles 0 holes 256..511 (256)

first mass with two stable charges: 36

charge holes below 84: [43, 61]

binding per nucleon peaks at A=62 (8.7946 MeV), the 6-bit layer

== F1 Stretch and fold: x -> {b x}; unwrapped it keeps everything, wrapped it shows one digit per tick ==

unwrapped 16^16 * pi, integer part in hex: 3243F6A8885A308D3 = the first 17 hex digits of pi, every crossing kept

wrapped {16x}, one crossing per tick: 243f6a8885a308d3 (the reader holds only the point on the circle)

== F2 The invertible form (baker's map): one direction expands forever, the other collapses forever ==

b= 2: 60 steps forward then 60 back returns exactly: True; y's first 60 base-2 digits = crossings newest-first: True

separation after 30 steps: x grew by b^30 exactly: True; y shrank by b^30 exactly: True; product unchanged: True

Lyapunov exponents +ln 2 = +0.6931 and -ln 2 = -0.6931: sum 0 (area preserved)

b=16: 60 steps forward then 60 back returns exactly: True; y's first 60 base-16 digits = crossings newest-first: True

separation after 30 steps: x grew by b^30 exactly: True; y shrank by b^30 exactly: True; product unchanged: True

Lyapunov exponents +ln 16 = +2.7726 and -ln 16 = -2.7726: sum 0 (area preserved)

== G1 Route back: in the complete transition graph the destination's own bits are the route ==

n=4: shifting in y's 4 bits lands on y from every x: True; longest shortest route = 4; pairs that need all 4 shifts: 74 of 256

n=8: shifting in y's 8 bits lands on y from every x: True; longest shortest route = 8; pairs that need all 8 shifts: 17622 of 65536

== G2 The readable universe is finite: horizon capacity under a cosmological constant ==

Lambda = 1.090e-52 m^-2, de Sitter horizon radius 1.659e+26 m, saturation density 1.3807e+69 bits/m^2

horizon capacity = 4.77e+122 bits (Lloyd's count of all operations so far ~1e120)

a fixed reader of k bits against an n-bit layer merges 2^(n-k) positions per reading: k=8, n=64 -> 2^56 = 7.206e+16

== G3 What every implementation agrees on: the entropy rate (Ornstein; Meshalkin's pair) ==

H(1/4,1/4,1/4,1/4) = 2.000000 bits H(1/2,1/8,1/8,1/8,1/8) = 2.000000 bits -> equal entropy, isomorphic shifts

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